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Relative age effect inverts when looking at career performance in elite youth academy soccer

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ABSTRACT

The aim of this study was to investigate the well-known selection bias favouring players born earlier in the year commonly referred to as the Relative Age Effect (RAE). Therefore, we analysed a group of top-tier youth academy players. The players joined and left the youth academy at different stages, where the first stage was at 8 years (U9) and the last stage was at 18 years of age (U19). Subsequently, we followed the career paths of all players in terms of minutes played in professional competitions. For that purpose, we collected competition information from transfermarkt.de. We label a competition as professional if the included teams obtain an average market value of 100.000€ (aggregated value of players playing in the team) and recorded the professional career minutes (PCM) for all players. Results show that in the youth academy, there are fewer players that are born late in the year compared to players that are born early in the year, confirming previous findings of RAE. However, we also find that players that are born late in the year achieve more PCM on average. This indicates that players that survive the RAE selection bias are exceptionally good at achieving long, successful careers.

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People show a general tendency to attribute success to individual merit, instead of situational circumstances (see e.g., Gilbert & Malone, 1995). However, a large body of research has shown that something as trivial as the date of a person's birth can have – under certain circumstances – a huge impact on an individual's career (Barnsley et al., 1985). For example, studies within professional sports show that the great majority of top-level athletes are born within the first months of the year, whereas far fewer players playing at the highest level are born later in the year (Cobley et al., 2009; Dudink, 1994; Edwards, 1994). This skewed birthdate distribution has been termed Relative Age Effect (RAE). Although the influence of the RAE in different sports is well established, particularly at the youth level (W. Helsen et al., 2005; Musch & Grondin, 2001; UEFA, 2016), initial studies have raised the possibility that a “relative age effect reversal” may exist in the long run. Hence, the present study builds on these initial findings by following the careers paths of top-tier German youth academy football players and analysing the minutes played in professional competitions.

The explanation of the RAE can be attributed to the fact that young athletes are divided into age-groups based on their birth date (usually with the cut-off date being the first of January) and those relatively older athletes may possess physical and cognitive advantages. Subsequently, this head start can result in a Matthew effect (“the rich get richer; the poor get poorer”, see e.g., Merton, 1968) due to more favourable circumstances, such as more coaching attention, more playing, and practice time. This line of argument is backed up by research suggesting that coaches have more favourable attitudes towards players of

advanced maturation (Furley & Memmert, 2015) and that maturation advantages have the potential to translate to performance advantages (Buchheit & Mendez-Villanueva, 2014; Gastin & Bennett, 2014), and that players have a higher market value (Furley et al., 2016). Hence, the most successful athletes are likely to show a skewed birthdate distribution (see e.g., W. F. Helsen et al., 2012).

Pertinent to the present research, studies have shown that the influence of relative age differences benefiting those born at the beginning of the year tends to decrease as players mature mentally and physically (Gibbs et al., 2012; McCarthy & Collins, 2014), and of particular importance, empirical evidence amongst more senior players (Ashworth & Heyndels, 2007; Coutts et al., 2014) suggests that players born towards the end of the year might show superior performance compared to their peers born earlier in the year. These findings have led to the hypothesis that athletes born in the later months of the year had to perform extraordinarily to progress through the talent development system in competitive sports (McCarthy & Collins, 2014). Hence, a growing body of studies is looking into the possibility of a RAE reversal by examining a range of performance indicators like salaries (Ashworth & Heyndels, 2007), first choices in player drafts (Baker & Logan, 2007), career length (Gibbs et al., 2012; Grossmann & Lames, 2013; Steingröver et al., 2016), games played and points scored (Deaner et al., 2013). Taken together, these studies suggest that, under certain conditions, athletes born at the end of the year (i.e., those that tended to be less often selected for teams in the younger categories) arrive at more senior-level categories (on average) with performance advantages compared to their peers born in

the earlier months. This pattern of results has been therefore termed the reversed RAE (RRAE).

A possible explanation for the RRAE might be that players born late in the year may have become highly skilled, because they were selected and maintained in the base categories, although they had physical and maturational disadvantages (Ashworth & Heyndels, 2007; Grossmann & Lames, 2013). More specifically, these players might have been driven to develop very high skill levels to overcome the difficulties of being among older players. This observation has paved the way for the underdog hypothesis, which contends that relatively younger players must possess or develop superior technical, tactical and psychological skills to compete with their relatively older counterparts (Cumming et al., 2018). In addition, these younger players might benefit from training with high proportions of relatively older players, which again might foster skill development. Indeed, Votteler and Höner (2014) have found advanced skill levels in relatively younger athletes, while Jackson and Comber (2020) have reported no evidence of RAE reversal. Additionally, some research (Deaner et al., 2013; Fumarco et al., 2017; Gibbs et al., 2012; Steingröver et al., 2016) provides evidence for performance advantages for players born relatively late in the year, whereas other research fails to find this effect (Coutts et al., 2014; McCarthy et al., 2015; Steingröver et al., 2016). These ambiguous findings call for further research on the reversal of the RAE and the possible implications for the underdog hypothesis.

The present research

The present research aimed to address some shortcomings of previous studies on the RRAE by investigating elite youth academy (EYA) football players in Germany over an extended period. All professional teams (1. & 2. Bundesliga) are obligated to run an elite youth academy (Höner et al., 2015). An EYA is an organizational unit that specializes in identifying, promoting, and training young talent in football. To achieve this, professional coaches work together with educators, physicians, and psychologists. EYA also enables its young athletes to take part in competitions at regional, national, and international levels in order to gain competitive experience and compete with other talented athletes.

One major strength of the present research is its longitudinal character (in contrast to cross-sectional analysis) as well as the long period and fine granularity of the dataset. The player-specific data reaches back to the season 2000/2001. This allows us to perform post-hoc evaluations of the careers of most players in the dataset. Hence, we do not need to predict or estimate career performance since the players already completed a large majority of their professional careers. This prevents us from possibly introducing an unknown bias from using predictive models. Instead, we can report accurate historic data that deterministically describes the career paths of the players in our dataset.

Admittedly, not all players in our dataset have ended their careers. For this reason, we excluded very young players as a large percentage of players in the remaining dataset is currently beyond 30 years of age (see specifications in Section

Sample) and has therefore already completed the majority of their career. Another strength of our dataset is its granularity. Since being recorded by the professional club itself, and not by a data provider, it contains all players who played in the youth academy at any stage of their career. This includes players who quit their career early before ever joining a professional team. In comparable datasets, these players are often missing since they do not have a record in publicly accessible databases like transfermarkt.de. This would introduce another bias that does not apply to our study.

Besides addressing shortcomings of previous research on the RRAE, the goal of the present research was to test for the RRAE effect for our sample of German football academy players. Thus, we would expect players born relatively late in the year to show a performance advantage in the success of their careers. As a proxy for career success, we chose professional career minutes (PCM) since they allow for a relatively position-agnostic comparison between players, whereas other metrics (such as monetary earnings, goals, assists, or clean sheets) are skewed towards different playing positions.

Method

Sample

Our sample consisted of a total of 1317 players. Inclusion criteria for a player were that he was officially registered as a youth player for an academy team at any point of a season. This information was gathered from an official list maintained by the youth academy that contained the player's name, the assigned youth team, and a link to the player's profile on transfermarkt.de (if one existed). Exclusion criteria were data errors in the official list and the birth date. For data errors, the official list contained 12 players with incomplete name information that restricted a reliable analysis. These players were excluded. For the birth date, we decided on a threshold of 16 years as a minimum age to collect PCM and excluded all players that were younger than this. As we performed our analysis in 2023, we excluded 322 players born after the 31st of December 2007.

After these preprocessing steps, we obtained a dataset of 1095 players. The average birth year of players was 1998 (± 16), while the oldest player was born in 1984. A total of 591 players were born before 1993, thus, being currently over 30 years of age. Of all players, 807 players did not play a single minute in a professional competition. The remaining 288 players played an average of 4898 PCM per player.

Procedure and measures

As previously explained, we computed PCM for all players using data from transfermarkt.de. Thus, for all players in our dataset, we obtained a unique transfermarkt.de ID that was manually checked to match the correct player identity. This ID was used to retrieve the career statistics of every player, collected by transfermarkt.de and publicly accessible over the transfermarkt.de site of the respective player. Player statistics are up-to-date until 8th of August 2023. After integration of the statistics from transfermarkt.de, we grouped players according to the month

that they were born in, as commonly performed by related studies on RAE (W. Helsen et al., 2005, 2012). Next, we iterate over all months and aggregate the PCM of all players born in that month. However, as this metric is obviously influenced by the number of players, which varies between different birth months, we divide the aggregated PCM by the number of players to obtain the mean PCM of a player born in that month.

Moreover, the aim of professional football players, generally, is to play at the highest professional level, which also correlates with higher monetary value. Thus, we needed to acknowledge the fact that not all minutes are equally valuable. In this respect, we looked at the possibility to introduce continuous weights to

different divisions, however, we discarded that possibility due to the connected degree of complexity that would be introduced by such a method. Instead, we decided to include all divisions above a hard threshold. We designed this threshold in discussions with a domain expert from the first team of the youth academy. Thus, we looked at different divisions where we found that a fair comparison between divisions can be established by using the mean market value (provided by transfermarkt.de) of players that are listed for teams in that division. Here, we do not have a problem with the position-specific bias in market value since the means for all teams are computed over all players (from all playing positions).

Table 1. Number of players, number of players with at least one PCM, aggregated PCM of all players born in that month, and mean and variance for PCM per player born in that month.

Month	Number of Players	Number of Professional Players	Total Aggregated PCM	Mean PCM per Player
Jan	194	46	128720	663.51
Feb	153	35	90600	592.16
Mar	144	31	156694	1088.15
Apr	117	29	108389	926.4
May	94	17	98305	1045.8
Jun	88	25	127144	1444.82
Jul	78	26	152373	1953.5
Aug	62	24	100533	1621.5
Sep	60	18	188705	3145.08
Oct	43	10	87813	2042.16
Nov	28	11	57229	2043.89
Dec	34	16	114194	3358.65

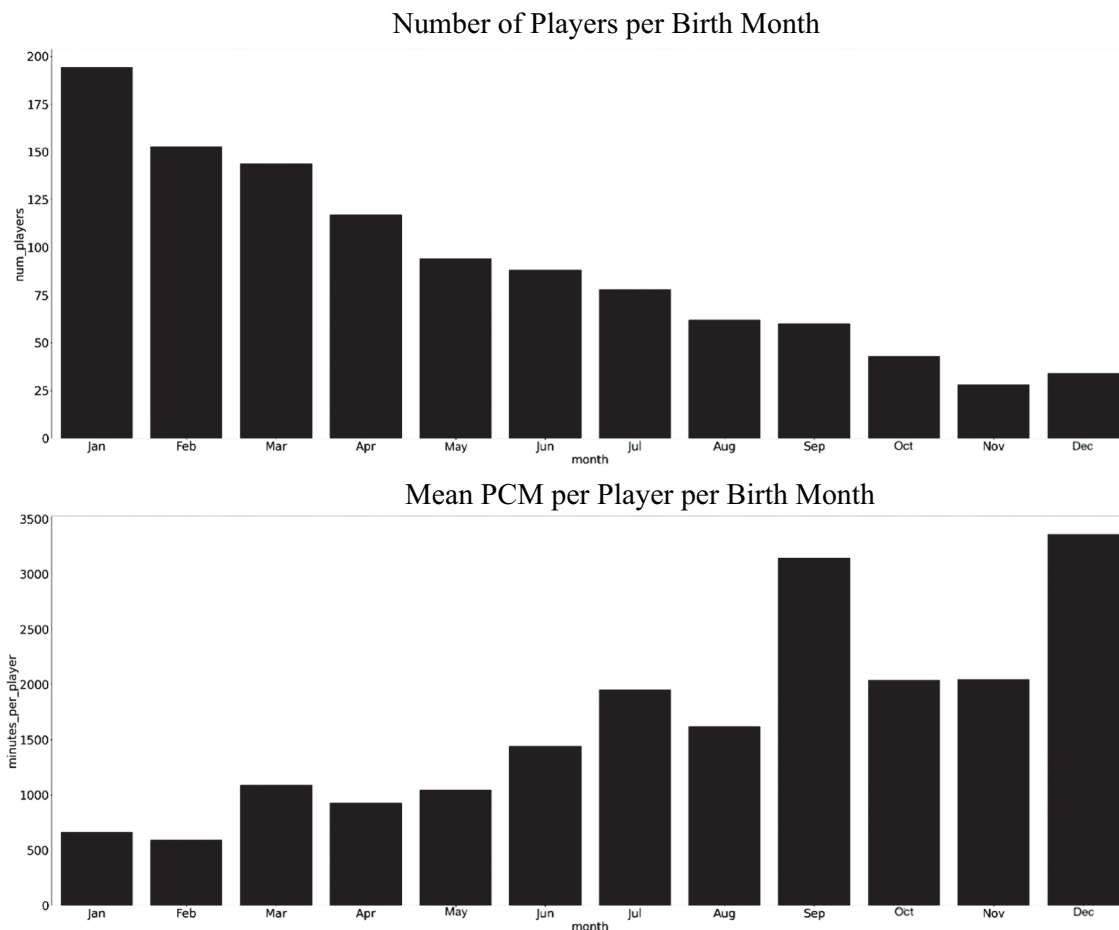


Figure 1. Total number of players and PCM per player visualized for every birth month. The RAE effect of decreasing player numbers over the year can be seen in the bottom plot. The reversal of the relative age effect (RRAE) is visible in the increasing mean PCM over the year.

Having decided on the metric to judge whether a league is professional or not, we still required a concrete value. Thus, we discussed different factors of professionalism such as team- and league infrastructure, player wages, crowd attendance, and others with our domain expert. Here, we found that a mean player market value of 100.000€ presents a good threshold between fully professional and semi-professional leagues.

Results and discussion

We present the number of players, the total aggregated PCM, and the mean in Table 1 and visualize these metrics in Figure 1.

Given the number of total players from every birth month in the academy (Figure 1, top), we were able to replicate the results of RAE, as reported in several data sets (Deaner et al., 2013; Fumarco et al., 2017; Gibbs et al., 2012; Steingröver et al., 2016). The highest number of 194 (17.7% of total) players was born in January. To test the RAE in our dataset we follow common procedure in the RAE literature and perform a chi-square test with the null hypothesis that the number of players is equally distributed along the four quartiles of the year (i.e., 25% per quartile). A total of 491 players (44.8%) were born in the first quartile, 299 (27.3%) were born in the second quartile, 200 (18.2%) were born in the third quartile, and 105 (9.6%) were born in the fourth quartile. Thus, we find a significant deviation from the expected uniform distribution (χ^2 [3, $N=1095$] = 298.633; $p < .001$, two-tailed; Cramer's $V = .302$; χ^2 [11, $N=1095$] = 322.721; $p < .001$; Cramer's $V = .164$, two-tailed for all 12 months), thus, rejecting the null-hypothesis and confirming the RAE for our dataset. Reasons for that effect have been largely attributed to the grouping mechanisms of youth academies, where players born earlier in the month are relatively older, and, thus, physically more advanced than players born later in that year (Buchheit & Mendez-Villanueva, 2014; Gastin & Bennett, 2014; Jackson & Comber, 2020).

While the potential ramifications of this selection bias have been discussed (Ashworth & Heyndels, 2007; Coutts et al., 2014), some players are selected by the youth academy despite being born late in the year. When comparing the number of players (Figure 1, top) with the mean PCM per player (Figure 1, bottom), it becomes visible how the RAE effect is reversed for career success in the average PCM. Thus, in contrast to the most players in the academy being born in January, the most PCM per player (3359) are achieved by players born in December. The descriptive statistics show that the average PCM per player linearly increases from early birth months to late months.

While players born in the first quartile average 781 PCM, players born in the second quartile already average 1139 PCM, players born in the third quartile average 2240 PCM, and players born in the fourth quartile average 2482 PCM. A 1×4 (birth quartile) univariate ANOVA revealed a significant main effect for birth quartile $F(3, 1091) = 7.302$, $p < .001$, $\eta^2_p = .020$. A polynomial linear contrast analysis confirmed a significant increase ($p < .001$) in average PCM as a function of birth quartile. This pattern of results was further confirmed by a regression analysis with birth month as a predictor for average PCM. This model could explain $R^2 = .021$. Birth month was a significant predictor ($\beta = .146$) of average PCM, $t = 4.886$, $p = .001$. Further analyses confirmed that it was significantly more likely—1.45 times more likely—to collect any PCM (as

compared to no-PCM) if a player was born in the second half of the year as compared to the first half of the year (χ^2 [1, $N=1095$] = 6.614; $p < .01$, two-tailed; Odds ratio = 1.451; 95% CI: 1.092–1.929). Stated differently, a player was one and a half times more likely to turn professional if they were born in the second half of the year. Thus, we reject the null hypothesis that PCM is equally distributed among players, and not influenced by their birth quartile. This was clearly not the case, and confirms the alternative hypothesis of the RRAE that players have more successful careers (in terms of PCM) the later they are born in the year.

Discussion and conclusion

Given ambiguous findings and methodological limitations of previous research on the RRAE, the goal of the present research was to test if EYA soccer players would accumulate more PCM during their career. We were able to demonstrate the RRAE in the domain of professional football and extend previous demonstrations of this effect (Ashworth & Heyndels, 2007; Baker & Logan, 2007; Deaner et al., 2013; Gibbs et al., 2012; Grossmann & Lames, 2013; Steingröver et al., 2016) in a long-term longitudinal data set. Players in our dataset, who were selected for youth academies even though being born later in the year, had longer and more successful professional careers, in terms of minutes played for professional teams. These results add to the recently emerged opinion in the literature showing that athletes who withstand the RAE selection bias in their youths perform extraordinarily as they mature in their careers. This effect most likely occurs as the players born in the later months of the year had to perform extraordinarily to progress through the talent development system in soccer and therefore became highly skilled. Yet, it must be noted that our study did not examine differences in the level of the professional competitions such that our observed effect might not translate to the highest level in soccer. In any case, the present research extends previous findings of the RAE in the German Elite Youth Academy System (Votteler & Höner, 2014) and is the first to show a reversal of the RAE in terms of career minutes.

The present research was able to address some remedies of previous research and contributes to the current debate on the RRAE. The strengths of our study are our long-term dataset that contains more details as datasets from related studies because it contains players that are usually not listed in public databases and contains information for more than 20 years. Thus, our results are valid and highly granular, as they are based on post-hoc career statistics and an evaluation metric derived from professional domain experts in football. Our findings are in line with Ashworth and Heyndels (2007) who reported that German soccer players born in the 4th quarter of the year tend to have higher salaries. Further, the present results support the findings of Deaner et al. (2013) and Fumarco et al. (2017) who analysed professional hockey players and found evidence for better performance of players born late in the year in terms of number of games, salaries, and the number of goals scored during their careers.

Our results can further be considered valuable for practitioners in adult and youth football to better understand the performances of players born late in the year. In this respect, the present findings may help improve talent detection and development as players

born towards the end of the year might tend to perform better in the long term, although they might have a performance disadvantage in earlier years. Yet, the implications of these results for the player selection process in youth academies need to be discussed. While different measures have been suggested to reduce the age bias in youth academies (Mann & van Ginneken, 2017), it is also true that the current age structure presents a challenging environment for relatively younger players. Therefore, a change in these circumstances might impact their ability to develop extraordinary skills. Further, it might also be valuable to focus on improving the circumstances that deter the development of the relatively older players in the youth academy (Cobley et al., 2009). Taken together, team managers and coaches might want to take all of these factors into consideration when selecting players for their teams.

In conclusion, the findings can be regarded as a further mosaic piece of evidence, pointing youth coaches towards focusing more on athletes' long-term skill development instead of short-term results (Grossmann & Lames, 2013; Höner et al., 2015; Memmert, 2015). Although, we acknowledge that further research is needed to scrutinize first findings on RRAE, we consider it likely that recruiting talented players who were cut from other teams due to relative disadvantages can (under certain circumstances) be beneficial for teams as these players may have a promising career ahead of them.

Disclosure statement

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Open science

All data will be made available at reasonable request to the corresponding author.

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